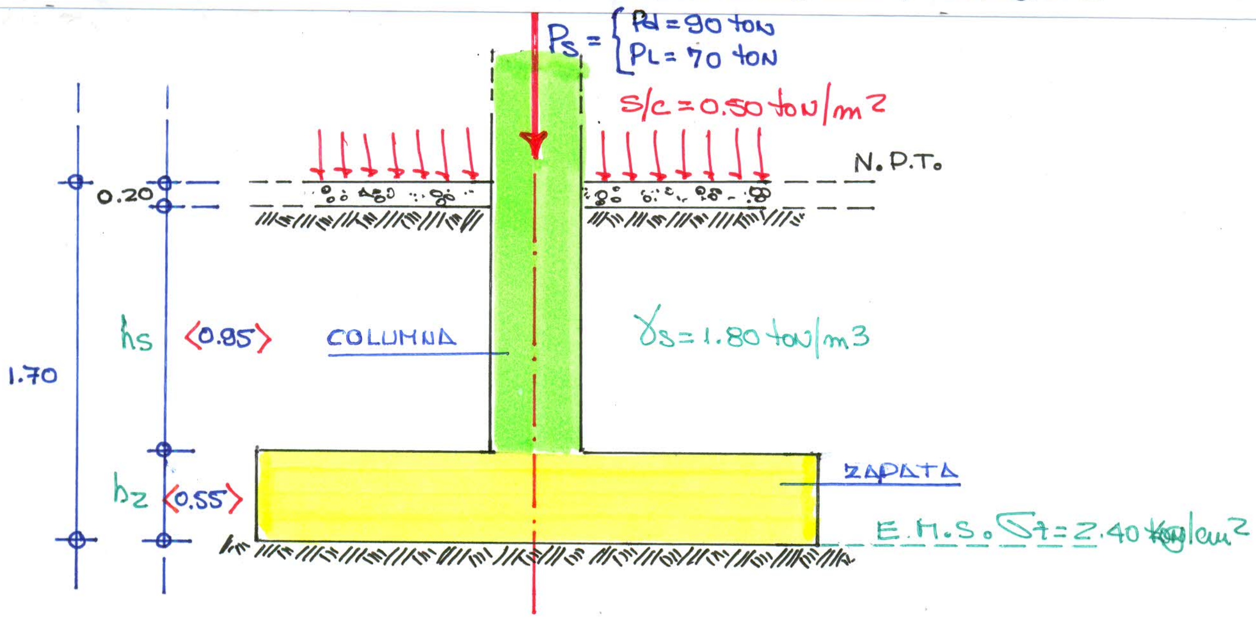
PROBLEMA N° 01:

Dimensionar, verificar y diseñar una zapata para soportar una columna de sección cuadrada de 0.40 m. de lado, que soporta una carga de Pd= 90 ton. , Pl= 70 ton., el nivel de fundación se encuentra a 1.70 m. por debajo del N.P.T, el suelo tiene un peso de 1.80 ton/m3, peso del piso que incluye sus acabados es de 1.90 ton/m3, sobrecarga de 0.50 ton/m2, la capacidad portante admisible del suelo es de 2.40 kg/cm2 según el estudio de mecánica de suelos, usar para el diseño concreto fc=280 kg/cm2, fy=4200 kg/cm2.



① DIMENSIONAMIENTO : ASUMIMOS $h = 0.55\,m. \Rightarrow d = h - 0.10 \Rightarrow d = 45\,cm.$

▸ CAPACIDAD EFECTIVA DEL SUELO (q_e) :

$$\boxed{q_e = \sigma_t - (\gamma c^o z * h_z) - (\gamma_s * h_s) - (\gamma c^o p * h_p) - s/c}$$

$$q_e = 24 - (2.40 * 0.55) - (1.80 * 0.95) - (1.90 * 0.20) - 0.50$$

$$q_e = 20.09\ ton/m^2$$

▸ AREA DE LA ZAPATA :

$$\boxed{A = \frac{P_s}{q_e}} \Rightarrow A = \frac{P_d + P_L}{q_e} \Rightarrow A = \frac{90+70}{20.09} \Rightarrow A = 7.964\ m^2$$

- DIMENSIONAMOS LA FORMA DE LA ZAPATA → f (FORMA DE LA COLUMNA)

$$L = \sqrt{A} \Rightarrow L = \sqrt{7.964} \Rightarrow L = 2.822\,m \longrightarrow L = 2.80\,m$$

② REACCION AMPLIFICADA DEL SUELO (q_u)

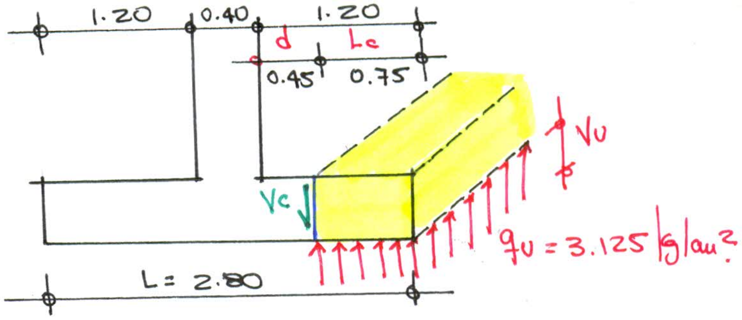
$$\boxed{q_u = P_u/A} \Rightarrow q_u = \frac{1.70\,P_L + 1.40\,P_d}{2.80 * 2.80} \Rightarrow q_u = 31.25\ ton/m^2$$

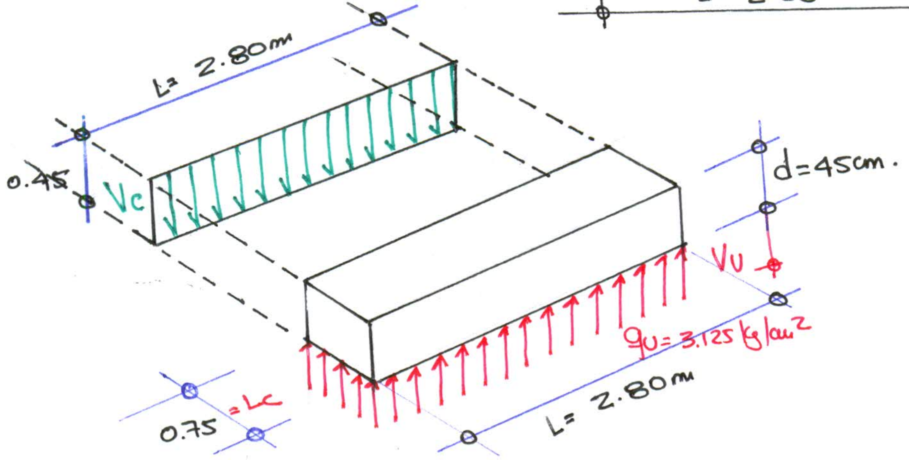
$$q_u = 3.125\ kg/cm^2$$

- VERIFICACIÓN POR CORTANTE.

$$\boxed{V_U \leq \phi V_c}$$

FUERZA CORTANTE ULTIMA ← → RESISTENCIA DEL CONCRETO @ CORTE





FUERZA CORTANTE ULTIMA V_U:

$$\boxed{V_U = q_U * \Delta}$$ AREA APLICADO

$$V_U = 3.125 * (75 * 280)$$

$$V_U = 65{,}625 \text{ kg} = 65.625 \text{ ton}$$

FUERZA DE RESISTENCIA DEL CONCRETO V_c

$$\boxed{\cancel{\phi} V_c = \cancel{\phi}\, 0.53 \sqrt{f'c}\; L\, d} \qquad \phi = 0.85$$

$$\phi V_c = 0.85 * 0.53 \sqrt{280} * 280 * 45$$

$$\phi V_c = 94{,}982.666 \text{ kg} = 94.983 \text{ ton}$$

$\therefore\ \underline{V_U < \phi V_c}$ OK CUMPLE

- PERALTE NECESARIO (d_{nec}):

$$V_U = \phi V_c$$

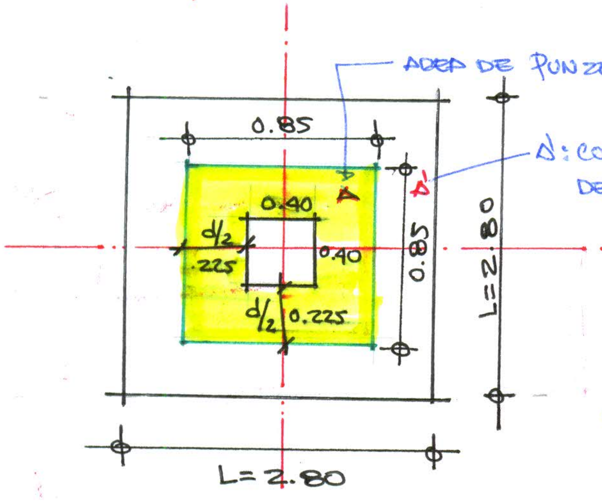
$$\boxed{q_U\, L_c * \cancel{L} = \phi 0.53 \sqrt{f'c} * \cancel{L} * d} \qquad d = d_{nec}$$

$$d_{nec} = \frac{q_U\, L_c}{\phi\, 0.53 \sqrt{f'c}} = \frac{3.125 * 75}{0.85 * 0.53 \sqrt{280}} \Rightarrow d_{nec} = 31.091 \text{ cm}$$

$d_{nec} < d_{asumido} = 0.45$ m.

VERIFICACION POR PUNZONAMIENTO:

$d = 45\ cm. \Rightarrow d/2 = 22.50\ cm.$

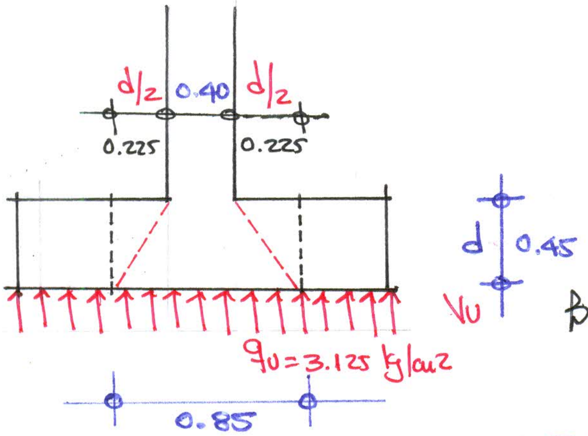


CONDICIÓN Q' DEBE CUMPLIR

$$\boxed{V_u \leq \phi V_c}$$

FUERZA CORTANTE ULTIMA

RESISTENCIA DEL CONCRETO AL CORTE



$A' = A_2 - A_{punz}$

$A' = 280^2 - 85^2$

$A' = 71,175.00\ cm^2$

PERIMETRO DE PUNZONAM.

$b_o = 4(85)$

$b_o = 340\ cm$

$\beta_o = \dfrac{\text{Lado > columna}}{\text{Lado < col.}} = \dfrac{0.40}{0.40} = 1$

CORTANTE MAXIMA PERMITIDA C°

$$\boxed{V_u = q_u\ A'}$$

$$\boxed{\phi V_c = \phi\left(0.53 + \frac{1.10}{\beta_o}\right)\sqrt{f'c}\cdot b_o\ d}$$

$V_u = 3.125 * 71,175.00$

$V_u = 222,421.875\ kg$

$= 222.422\ ton$

$\phi V_c = 0.85\left(0.53 + \frac{1.10}{1}\right)\sqrt{280}\ 340\ 45$

$\phi V_c = 354712.895\ kg$

$= 354.713\ ton$

$\therefore \boxed{V_u < \phi V_c < V_{cmax}}$ CUMPLE LA CONDICION.

PERALTE NECESARIO:

CORTE ACTUANTE — CORTE RESISTENTE MAX C°

$V_u = q_u * A' \quad = \quad V_c = v_c\ b_o d$

$v_c = \phi\ 1.10\sqrt{f'c}\ (1 \times 1)$

$v_c = 0.85 \times 1.10\sqrt{280}$

$v_c = 15.646\ kg$

$V_u = V_c$

$q_u * A' = v_c\ b_o d \qquad d = d_{mec}$

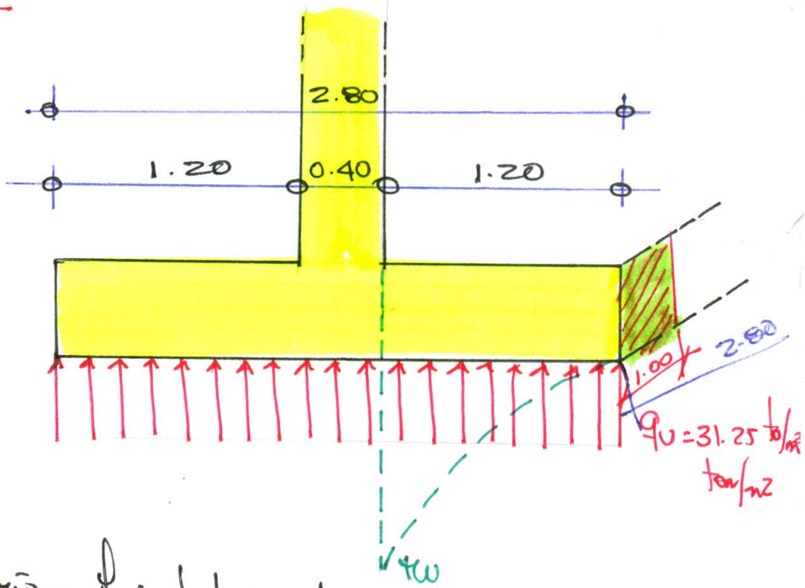
$$\boxed{d_{mec} = \frac{q_u * A'}{v_c\ b_o}} \Rightarrow d_{mec} = \frac{3,125 * 71,175.00}{15.646 * 340\ cm} \Rightarrow d_{mec} = 41.811\ cm$$

$\therefore d_{\text{DISEÑO}} = 45\ cm. > d_{mec} = 41.811\ cm$ OK

Diseño de Refuerzo por Flexión

$$Mu = \frac{q_u \cdot L^2}{2}(1.00)$$

$$Mu = \frac{31.25 \times 1.20^2}{2} \Rightarrow Mu = 22.50\ ton\text{-}m$$



Datos de Diseño:

$f'c = 280\ kg/cm^2, \quad f_y = 4200\ kg/cm^2$

$d = 45\ cm. \quad b = 100\ cm.$

Parámetros de Diseño

$$\rho_{min} = \frac{0.70\sqrt{f'c}}{f_y} = 0.00279 \qquad As_{min} = \rho_{min}\, b d \Rightarrow As_{min} = 12.5499\ cm^2$$

$$\rho_b = 0.85\beta_1 \frac{f'c}{f_y}\left(\frac{6000}{6000+f_y}\right) = 0.02833 \qquad As_b = \rho_b\, bd \Rightarrow As_b = 127.50\ cm^2$$

$$\rho_{max} = \rho_b(0.75) = 0.02125 \qquad As_{max} = \rho_{max}\, bd \Rightarrow As_{max} = 95.625\ cm^2$$

$$Mu_{max} = \phi As_{max} f_y\left(d - \frac{a_{max}}{2}\right) \qquad Mu_{max} = 132.159\ ton\text{-}m > Mu$$

$$a_{max} = \frac{As\, f_y}{0.85 f'c\, b} \Rightarrow 16.875 \qquad \therefore\ Mu < Mu_{max} \Rightarrow D.S.R.$$

$$\boxed{0.59\omega^2 - \omega + \frac{Mu}{\phi b d^2 f'c} = 0}$$

$$\frac{22.50 \times 10^5}{0.90 * 45^2 * 100 * 280} = 0.04409$$

$$\rho_d < \rho_b \quad \text{Falla dúctil}$$

$\therefore\ 0.59\omega^2 - \omega + 0.04409 = 0$

$$\begin{cases} \omega_1 = 1.64961 \\ \omega_2 = 0.04530 \end{cases} \Rightarrow \omega = 0.0453$$

① $\boxed{\omega = \rho_d \frac{f_y}{f'c}} \Rightarrow \boxed{\rho_d = \omega \frac{f'c}{f_y}} \Rightarrow \rho_d = \frac{0.0453(280)}{4200} \Rightarrow \rho_d = 0.00302$

② $\boxed{As_d = \rho_d\, bd} \Rightarrow As_d = 0.00302(100)(45) \Rightarrow As_d = 13.59\ cm^2 \Rightarrow As_{dt} = 38.052\ cm^2$ $*2.80$

$$\boxed{n = \frac{As_{dt}}{\phi_{As}}} \qquad \phi_{As} = 5/8'' = 2.00\ cm^2 \Rightarrow n = \frac{38.052}{2.00} \Rightarrow n = 19.026 \sim n = 19\ \text{und}$$

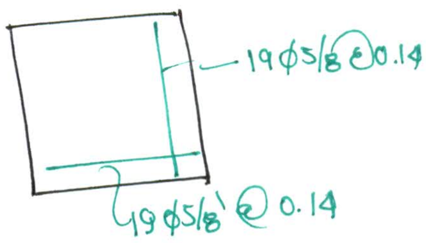
Diámetro

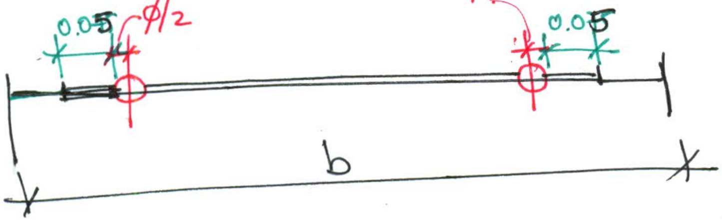
$\phi 3/4 = 0.019$

$\phi 5/8 = 0.016$

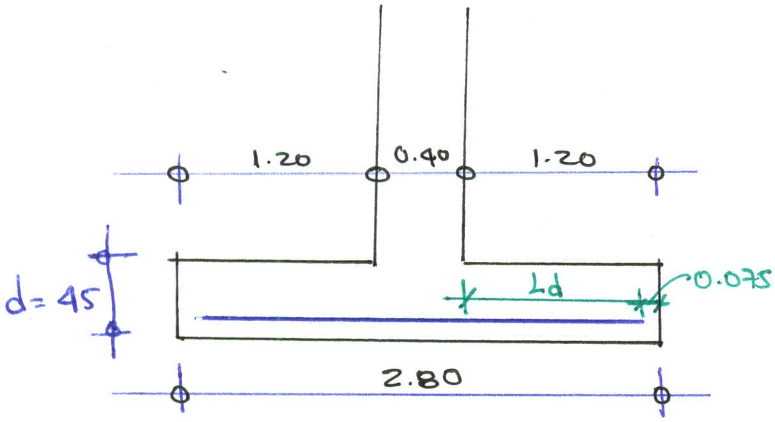
$$\boxed{S = \frac{b - 2r - \phi - 0.10}{n-1}} \Rightarrow S = \frac{2.80 - 2(0.075) - 0.016 - 0.10}{19-1} \Rightarrow S = 0.141\ m.$$

19 ϕ5/8" @ 0.14 m Ambos Sentidos





LONGITUD DE DESARROLLO DEL REFUERZO:



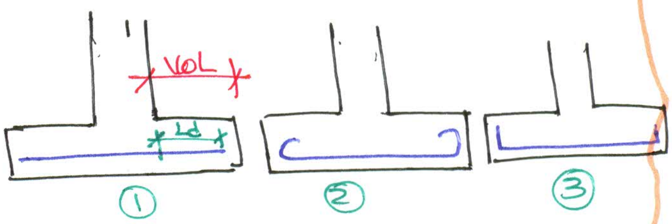
NORMA: LONG. DESARROLLO TRACCIÓN

$Ld = 0.06 \frac{Ab\, fy}{\sqrt{fc}}$

$Ld = 0.006\, db\, fy$

$Ld \geq 30cm$

USAR EL MAYOR



$Vol \geq Ld \Rightarrow$ RECTO (1)

$Vol < Ld \Rightarrow$ DOBLADO ó GANCHO (2 y 3)

LONGITUD DE BARRA: (Lb)

$Lb = Lv - r \Rightarrow Lb = 1.20 - 0.075 \Rightarrow Lb = 1.125\ m$

LONGITUD DESARROLLO BARRA BÁSICA (Ldb)

$Ab\ \phi 5/8 = 2.00\ cm^2$

$db\ \phi 5/8 = 0.016\ m.$ 1.588 cm.

$Ldb = 0.06\ Ab\ fy/\sqrt{fc} = 0.06\ (2.00)4200/\sqrt{280} \Rightarrow Ldb = 30.12\ cm.$ ✗

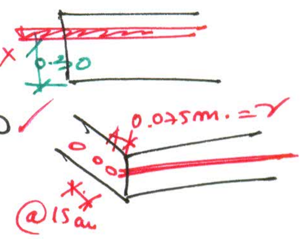
$Ldb = 0.006\ db\ fy = 0.006\ (1.588)4200 \Rightarrow Ldb = 40.018\ cm.$ ✓ $Ldb = 40\ cm$

$Ldb \geq 30\ cm = 30$ $Ldb \geq 30.00$ ✗

LONGITUD DE DESARROLLO: (Ld) - - - - - - $\boxed{Ld = Ldb\ (\text{Factor})}$ NORMA.

$Ld = 0.80\ Ldb \Rightarrow Ld = 0.80(40) \Rightarrow Ld = 32.00\ cm$

Factor { 1.40 ✗ / 0.80 ✓



$\Rightarrow Ld = 0.32m < Lb = 1.125m$ ⟨CONFORME⟩

$\therefore Lvol = 1.20m > Ld = 0.32m$ ACERO RECTO SIN GANCHO:

TRANSFERENCIA DE FUERZAS EN EL INTERFASE (COLUMNA - CIMENTACIÓN)

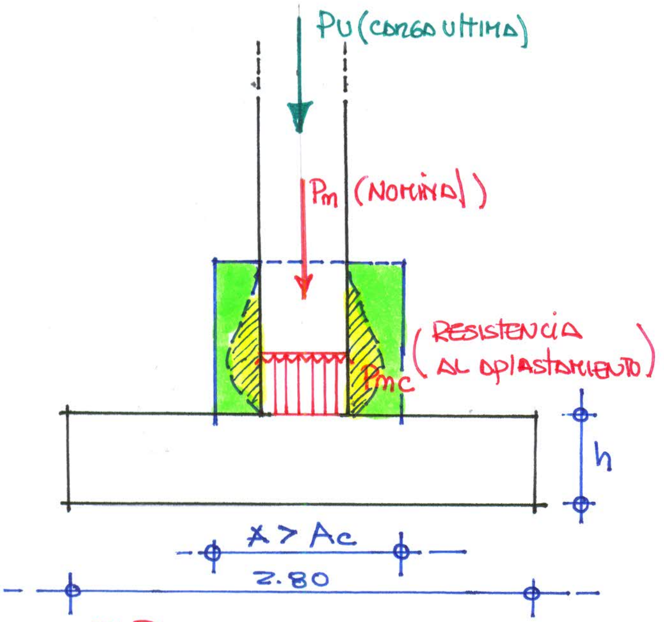
DATOS: $f'c = 280\ kg/cm^2$: $P_U = 1.70\ P_L + 1.40\ P_d$

$P_d = 90$ ton $\quad P_U = 1.70(70) + 1.40(90) \Rightarrow \underline{P_U = 245\ ton}$

$P_L = 70$ ton

: MÍNIMO $\phi P_m = P_U \Rightarrow P_m = \dfrac{P_U}{\phi} \Rightarrow P_m = \dfrac{245}{0.70} \Rightarrow \underline{P_m = 350\ ton}$

A) RESISTENCIA AL APLASTAMIENTO DEL CONCRETO SOBRE LA COLUMNA P_{mc}



$\boxed{P_{mc} = 0.85\ f'c\ A_c} \Rightarrow P_{mc} = 0.85 * 280 * 40^2$

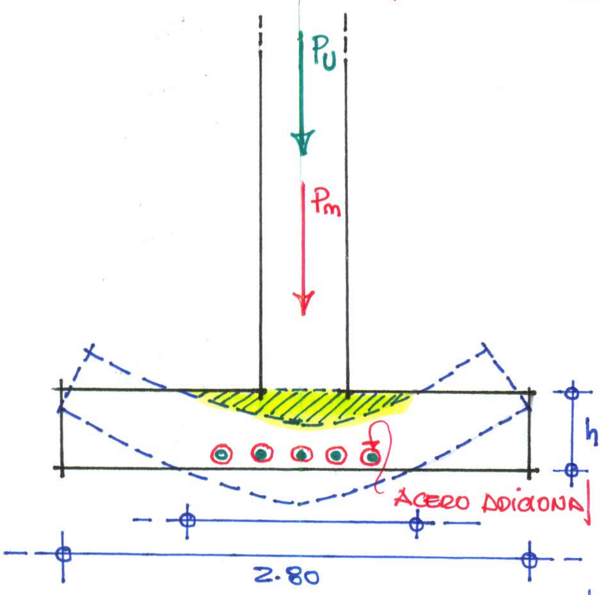
$\underline{P_{mc} = 380800\ kg = 380.80\ ton}$

$\boxed{P_{mc} > P_m}$ No REQUIERE PEDESTAL

$P_{mc} = 380.80$ ton $> P_m = 350.00$ ton

RESISTENCIA COL > CARGA NOMINAL (OK)

B) RESISTENCIA AL APLASTAMIENTO DEL CONCRETO EN LA CIMENTACIÓN (P_{mz})



$\boxed{P_{mz} = 0.85\ f'c\ A_o}$ $\quad A_o = \sqrt{\dfrac{A_z}{A_c}} \leq 2\ col$

SI $\sqrt{\dfrac{A_z}{A_c}}$ $> 2 \Rightarrow A_o = 2A_c$; $\leq 2 \Rightarrow A_o = 2$

$\sqrt{\dfrac{A_z}{A_c}} = \sqrt{\dfrac{2.80^2}{0.40^2}} = 7.00 > 2 \Rightarrow A_o = 2\ A_c$

$A_o = 2(0.40^2)$

$A_o = 0.32\ m^2$

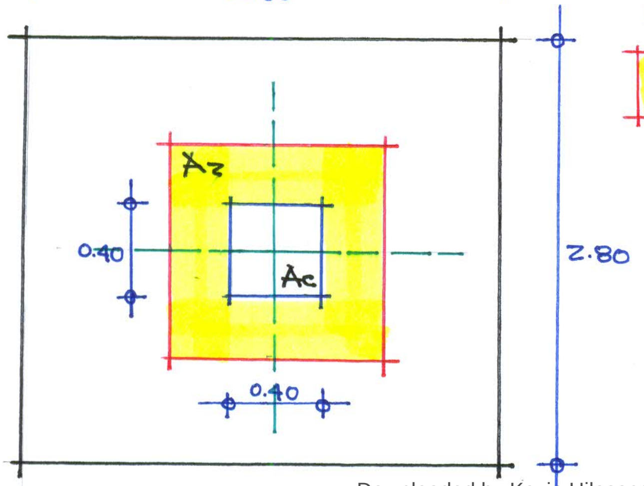
$P_{mz} = 0.85\ f'c\ A_o$

$P_{mz} = 0.85 * 280 * 2(40^2) \Rightarrow P_{mz} = 761\ 600\ kg$

$P_{mz} = 761.60$ ton $> P_m = 350$ ton

RESIST. ZAPATA > CARGA ULTIMA NOMINAL

$\boxed{P_{mz} > P_m}$ No REQUIERE REF. ADICIONAL

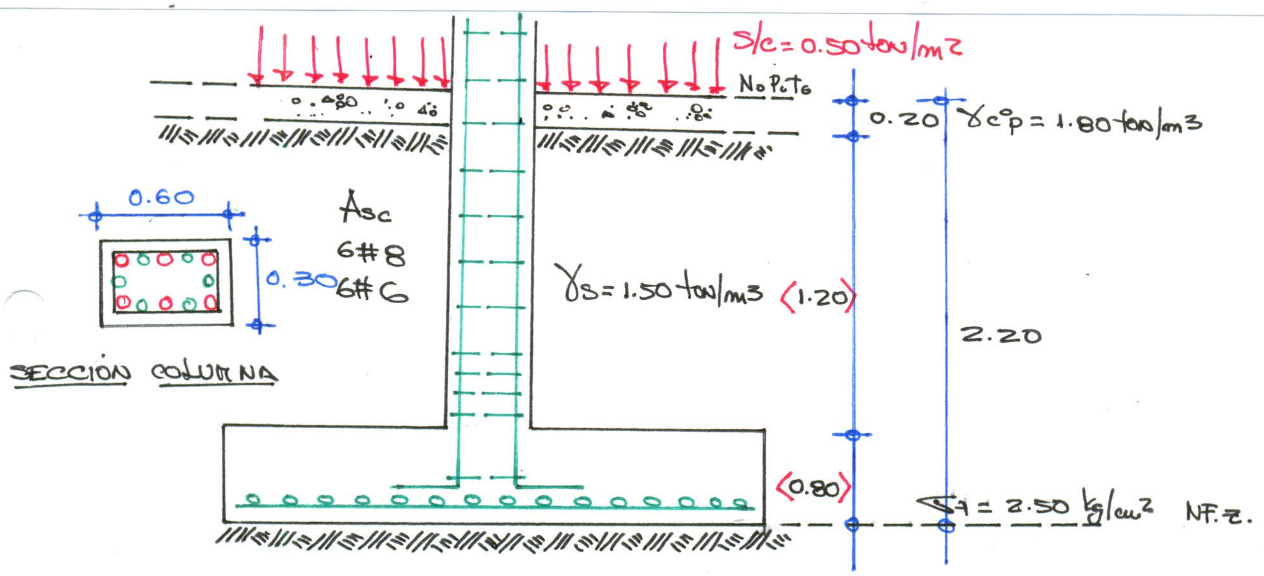


A_z = ÁREA DE SUPERFICIE DE APOYO

A_c = ÁREA CARGADA

PROBLEMA N° 02:

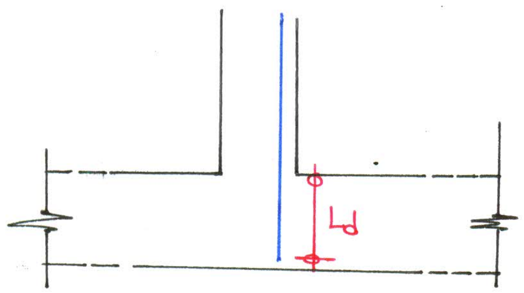
Determinar, verificar y diseñar la zapata aislada mostrada en la figura adjunta que soporta una columna de sección rectangular de 0.30x0.60 m. (bxt) reforzada con 6 varillas # 8 y 6 varillas # 6, la carga que transmite a la columna es de Pd=180 ton, Pl=120 ton., el peso específico del suelo es de 1.50 ton/m3 y la capacidad portante admisible del suelo según el (EMS) es de 2.50 kg/cm2, además está sometido a la acción de una sobrecarga de 500 kg/m2, emplear para el diseño fc=210 kg/cm2 y fy=4200 kg/cm2.



I DIMENSIONAMIENTO DE LA ZAPATA:

$$A_{sCOL}\begin{cases}6\#8 = 1'' \\ 6\#6 = 3/4''\end{cases} \Rightarrow \text{ELEGIMOS EL MAYOR } \#8 = 1'' \begin{cases}A_b = 5.10\ cm^2 \\ d_b = 2.54\ cm.\end{cases}$$

"d" DEBE SER CAPAZ PERMITIR EL DESARROLLO DEL REFUERZO EN COMPRESIÓN EN LA COLUMNA



∴ NORMAS PERUANAS Ld COMPRESIÓN
$$\begin{cases}L_d = 0.08\ d_b\ f_y/\sqrt{f'c} \\ L_d = 0.004\ d_b\ f_y \\ L_d \geq 20\ cm\end{cases}$$

$$L_d = 0.08 d_b\ f'_y/\sqrt{f'c} = 0.08\,(2.54)\,4200/\sqrt{210} = 58.893\ cm.$$

$$L_d = 0.004\ d_b\ f_y = 0.004\,(2.54)\,4200 = 42.67\ cm \Rightarrow L_d = 60\ cm.$$

$$L_d \geq 20\ cm = 20\ cm. = 20.00\ cm.$$

$$\therefore d = L_d + 0.10 \Rightarrow d = 70\ cm. \Rightarrow h = d + \underbrace{r + \phi}_{0.10} \Rightarrow h = 80\ cm.$$

CAPACIDAD EFECTIVA DEL TERRENO = CAPACIDAD PORTANTE NETA DEL TERRENO (q_e)

$$\boxed{q_e = \sigma_t - (\gamma_{c^\circ z}\, h_z) - (\gamma_s\, h_s) - (\gamma_{c^\circ p}\, h_p) - s/c}$$

$q_e = 25 - (2.40 * 0.80) - (1.50 * 1.20) - (1.80 * 0.20) - 0.50$ .... $\Rightarrow q_e = 20.42$ ton/m²

$q_e = 2.042$ kg/cm²

AREA DE LA ZAPATA REQUERIDA : $q_e = 20.42$ ton/m², $P_d = 180$ ton, $P_L = 120$ ton.

$$\boxed{A = \frac{P_T}{q_e}} \quad A = \frac{P_d + P_L}{q_e} = \frac{180+120}{20.42} \Rightarrow A = 14.691\ m^2$$

DIMENSIONAMIENTO DE LOS LADOS DE LA ZAPATA :

$$\boxed{A_z = (b+2m)(t+2m)}$$

$14.691 = (0.30+2m)(0.60+2m)$

$14.691 = 0.180 + 0.60m + 1.20m + 4m^2$

$4m^2 + 1.80m - 14.511 = 0$

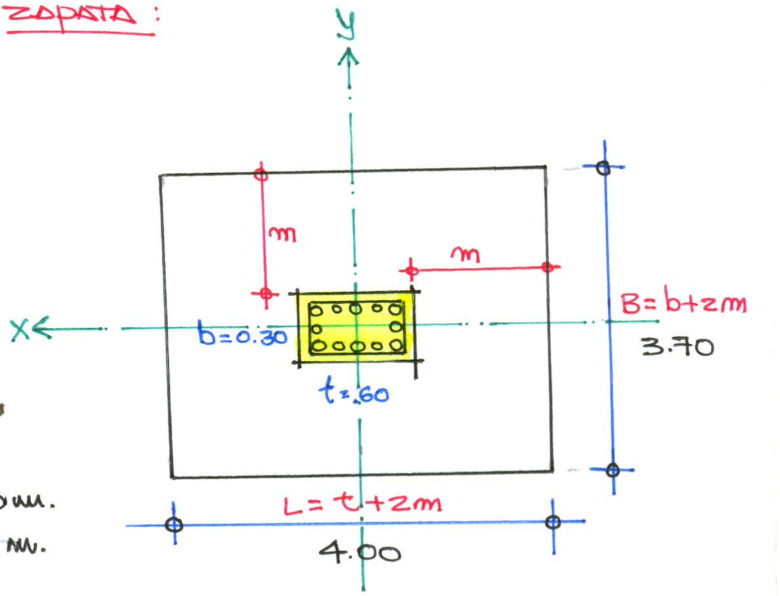
$m_1 = 1.693\,m.$

$m_2 = -2.143\,m \quad \therefore \; m = 1.70\,m$

$B = b+2m = 0.30 + (1.70)2 \Rightarrow B = 3.70\,m.$

$L = t+2m = 0.60 + (1.70)2 \Rightarrow L = 4.00\,m.$

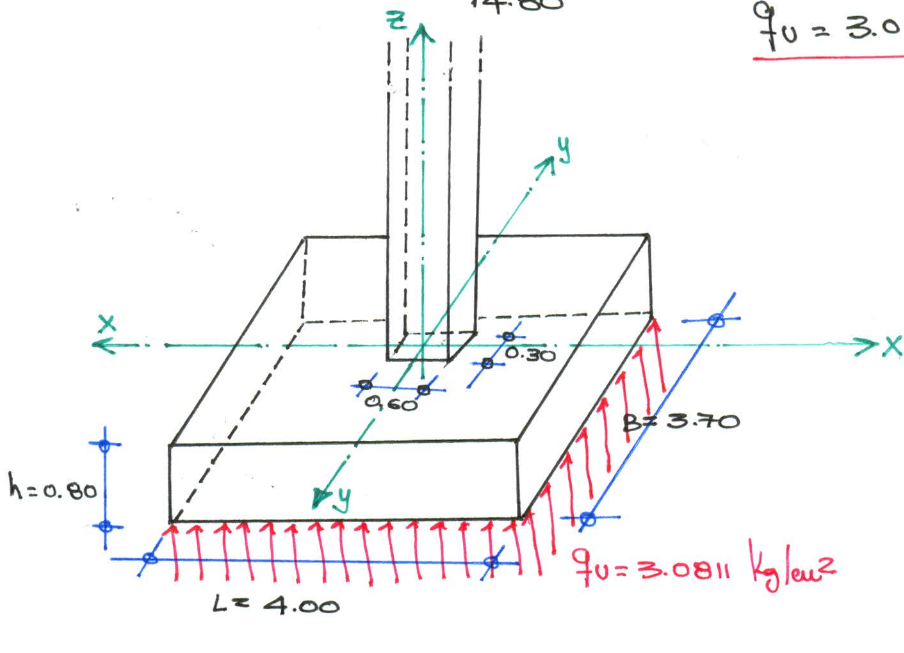
$\Rightarrow A_z = 14.80\ m^2$



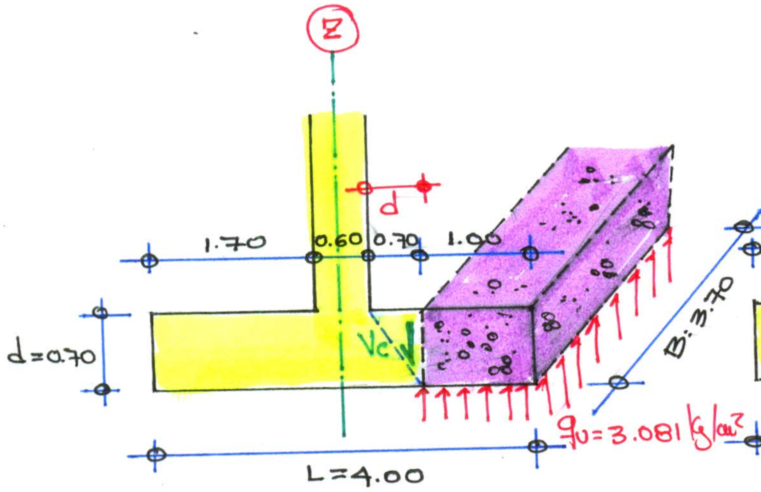
REACCION AMPLIFICADA DEL SUELO (q_u) SEGUN RNE : $P_u = 1.70\,P_L + 1.40\,P_d$

$$\boxed{q_u = \frac{P_u}{A}} = \frac{1.70\,P_L + 1.40\,P_d}{A} = \frac{1.70(120) + 1.40(180)}{14.80} \Rightarrow q_u = 30.811\ \text{ton/m}^2$$

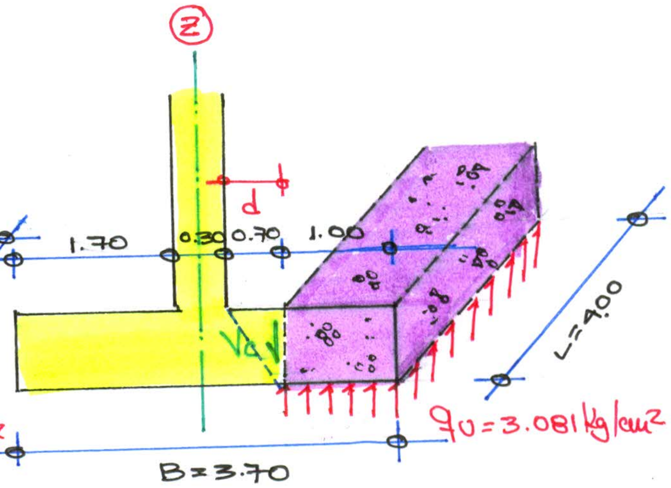
$q_u = 3.0811$ kg/cm²



Verificacion por corte: flexion



DIRECCION: X-X



DIRECCION: Y-Y

Condicion Normativa

$$\boxed{V_u \leq \phi V_c}$$

Fuerza cortante ultima — Resistencia del concreto @ corte

Dirección X-X

$$\boxed{V_u = q_u * A}$$

$V_u = 3.081 * (1.00 * 3.70)$

$V_u = 113\,997.00\ kg = 113.997\ ton$

$$\boxed{\phi V_c = \phi\, 0.53 \sqrt{f'c} * B * d}$$

$\phi V_c = 0.85 * 0.53 \sqrt{210} * 370 * 70$

$\phi V_c = 169\,084.659\ kg = 169.084\ ton$

$V_u = 113.997\ ton < \phi V_c = 169.084\ ton$

Conforme

Dirección Y-Y

$$\boxed{V_u = q_u * A}$$

$V_u = 3.081\ (100 * 400)$

$V_u = 123\,240.00\ kg = 123.240\ ton$

$$\boxed{\phi V_c = \phi\, 0.53 \sqrt{f'c}\ L * d}$$

$\phi V_c = 0.85 * 0.53 \sqrt{210} * 400 * 70$

$\phi V_c = 182\,794.226\ kg = 182.794\ ton$

$V_u = 123.240 < \phi V_c = 182.794\ ton$

Conforme

Peralte Necesario (d_{mec})

$q_u A = \phi\, 0.53 \sqrt{fc}\ L\, d$

$l_c = 100\ cm.$

$q_u (L_c * \cancel{L}) = \phi 0.53 \sqrt{fc}\ \cancel{L}\, d$

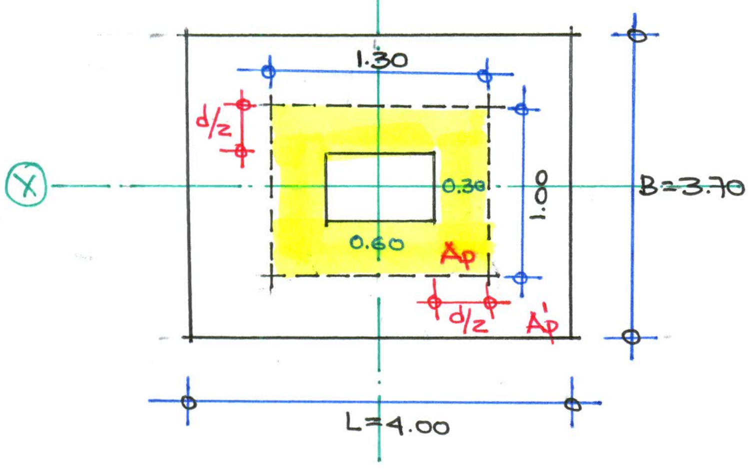
$q_u\, l_c = \phi\, 0.53 \sqrt{fc}\ \underline{d}$

$$\boxed{d_{mec} = \frac{q_u\, l_c}{\phi 0.53 \sqrt{fc}}} \Rightarrow d_{mec} = \frac{3.081 * 100}{0.85 * 0.53 \sqrt{210}} \Rightarrow d_{mec} = 47.194\ cm$$

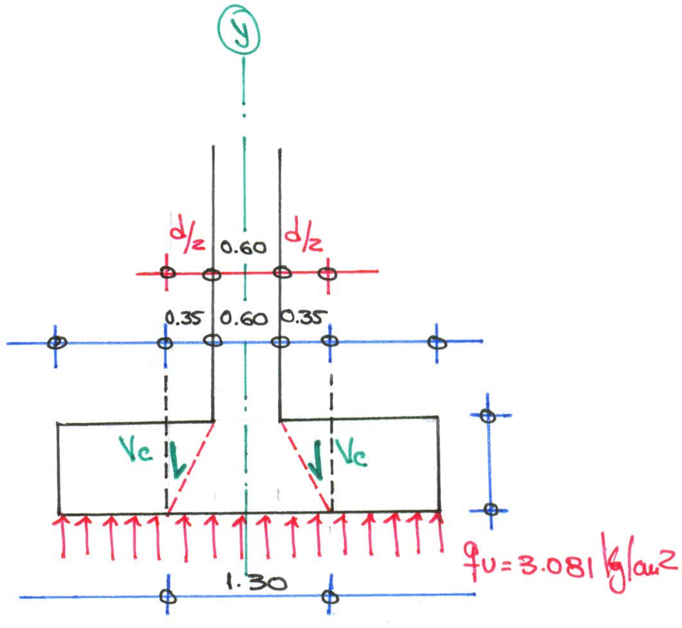
$d_{mec} = 4\ldots < d_{\ldots} = 70.00\ cm$

Conforme

VERIFICACION POR PUNZONAMIENTO:



PLANTA



SECCION: X – X

CONDICION NORMATIVA

Fuerza cortante ultima $\boxed{V_u \leq \phi V_c}$ Resistencia del concreto @ corte

$$A'p = A_Z - A_P$$

$$A'p = (400 * 370) - (130 * 100)$$

$$A'p = 135\,000 \text{ cm}^2$$

$$\beta_o = \frac{\text{lado} > \text{col.}}{\text{lado} < \text{col.}} = \frac{0.60}{0.30} \Rightarrow \beta_o = 2$$

Perimetro de Punzonamiento

$$b_o = 2(100 + 130)$$

$$b_o = 460.00 \text{ cm.}$$

$$V_u = q_u * A'p$$

$$V_u = 3.081 * 135000$$

$$V_u = 415\,935.00 \text{ kg} = 415.935 \text{ ton}$$

$$\phi V_c = \phi\left(0.53 + \frac{1.10}{\beta_o}\right)\sqrt{f'c}\; b_o d$$

$$\phi V_c = 0.85\left(0.53 + \frac{1.10}{2}\right)\sqrt{210} * 460 * 70$$

$$\phi V_c = 428\,359.30 \text{ kg} = 428.359 \text{ ton}$$

$$\therefore V_u = 415.935 \text{ ton} < \phi V_c = 428.359 \text{ ton}$$

CONFORME

PERALTE NECESARIO d_{nec}

$$\boxed{v_o b_o d = q_u A'p}$$

$$\boxed{v_o = \phi\, 1.10\sqrt{210}}$$

$$v_o = 0.85 * 1.10\sqrt{210}$$

$$v_o = 13.549 \text{ kg.}$$

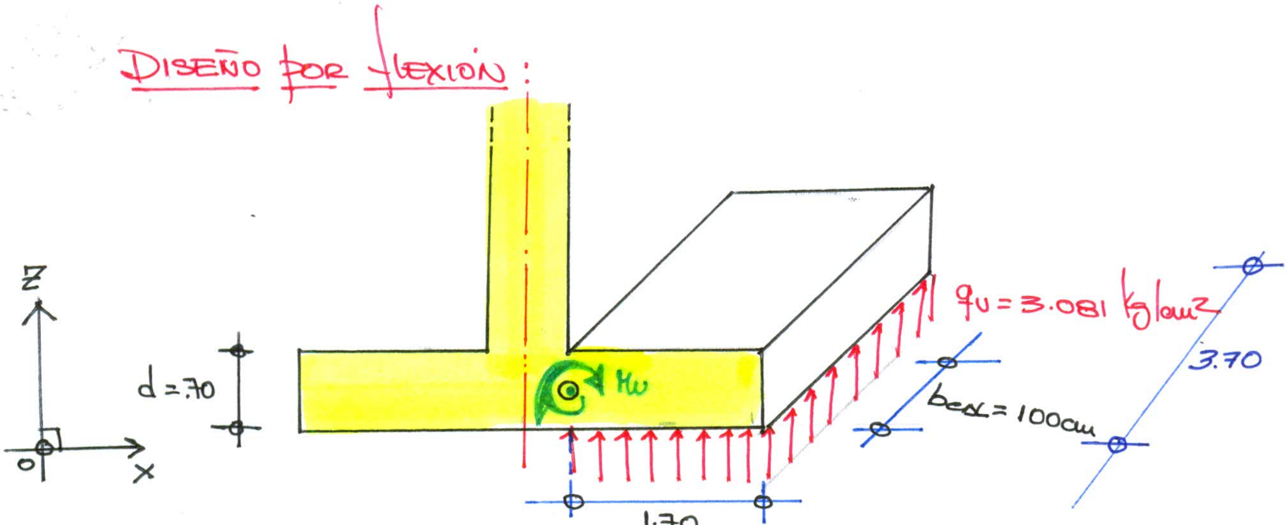
$$d_{nec} = \frac{q_u A'p}{v_o b_o} = \frac{3.081(135000)}{13.549 * 460}$$

$$d_{nec} = 66.736 \text{ cm}$$

$$\therefore d_{nec} = 66.736 \text{ cm} < d_{col} = 70.00 \text{ cm}$$

CONFORME

DISEÑO POR FLEXIÓN:



SECCIÓN: X-X

$\boxed{Mu = \frac{qu\,L^2}{2} b_{eol}} \Rightarrow Mu = \frac{30.81\,(1.70)^2\,(1.00)}{2} \Rightarrow Mu = 44.52 \text{ ton-m}$

DATOS DE DISEÑO

$b = 100 cm$

$d = 70 cm$

$f'c = 210 \text{ kg/cm2}$

$f'y = 4200 \text{ kg/cm2}$

$Mu = 44.52 \text{ ton-m.}$

$\boxed{a = \frac{As\,fy}{0.85\,f'c\,b}}$ $\boxed{Mu = \phi\,As\,fy\,(d - \frac{a}{2})}$

$0.59\,\omega^2 - \omega + \frac{Mu}{\phi b d^2 f'c} = 0$

$\omega = \rho_d \frac{fy}{f'c}$, $\rho_d = Asd/bd$, $\rho_d = \omega \frac{f'c}{fy}$, $Asd = \rho_d\, b d$

$\rho_{min} = \frac{0.70\sqrt{f'c}}{fy} \Rightarrow \rho_{min} = 0.00242$ $\quad As_{min} = \rho_{min}\, bd \Rightarrow As_{min} = 16.90661 cm^2$

$\rho_b = 0.85 \beta_1 \frac{f'c}{fy}\left(\frac{6000}{6000+fy}\right) \Rightarrow \rho_b = 0.02125$ $\quad Asb = \rho_b\, b d \Rightarrow Asb = 148.75 cm^2$

$\rho_d = \quad \Rightarrow \rho_d = 0.00248$ $\quad Asd = \rho_d\, bd \Rightarrow Asd = 17.33177 cm^2$

$\rho_{max} = 0.75\,\rho_b \Rightarrow \rho_{max} = 0.01594$ $\quad As_{max} = \rho_{max}\, bd \Rightarrow As_{max} = 111.562 cm^2$

$\rho_d < \rho_{max} < \rho_b$ ∴ FALLA DUCTIL

DISTRIBUCIÓN DE ACEROS:

$Astd = 17.332 * 3.70 \Rightarrow Astd = 64.128 cm^2$

$\phi 3/4'' = 2.84 cm^2$

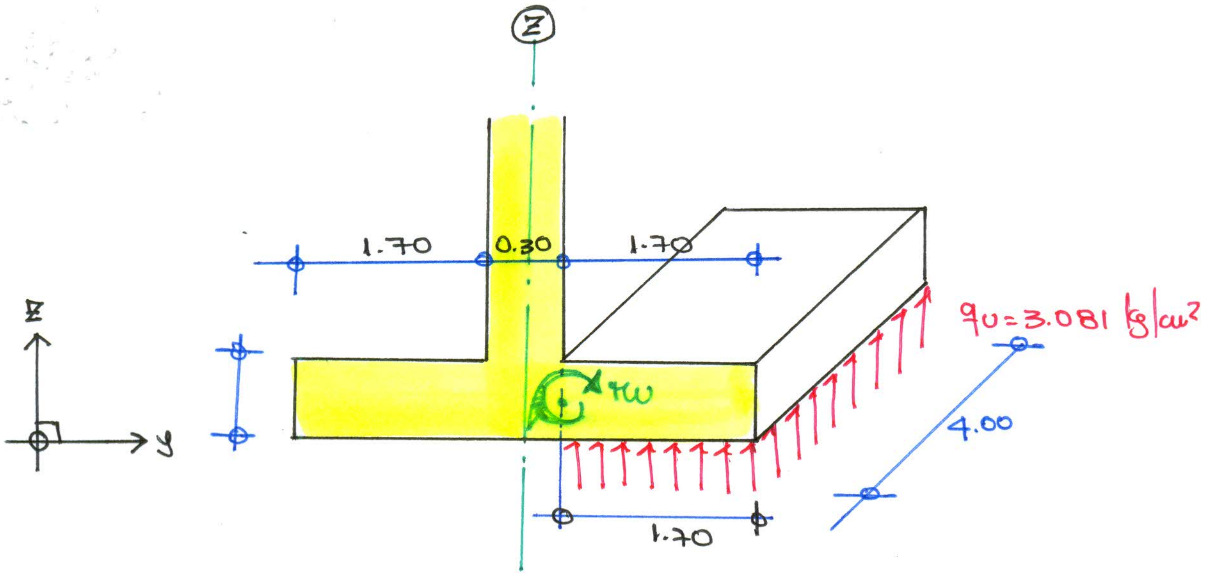
$\boxed{m = \frac{Astd}{As\phi}} \Rightarrow m = \frac{64.128}{2.84} \Rightarrow m = 22.58 \Rightarrow m = 23$ UNIDADES

ESPACIAMIENTO:

$db\ \phi 3/4 = 0.019 m$

$\boxed{S = \frac{B - 2(r) - \phi - 0.10}{m-1}} = \frac{3.70(2*0.075) - 0.019 - 0.10}{22} \Rightarrow S = 0.156 cm.$

$23\ \phi 3/4 @ 0.156 cm$:



SECCIÓN : y-y

$\boxed{Mu = \dfrac{q_u L^2 \, b_{col}}{2}} \Rightarrow Mu = \dfrac{3.081 * 1.70^2 x 100}{2} \Rightarrow Mu = 44.52 \text{ ton-m}$

DATOS DE DISEÑO:

$b = 100 \text{ cm}$

$d = 70 \text{ cm}$

$f'c = 210 \text{ kg/cm}^2$

$fy = 4200 \text{ kg/cm}^2$

$Mu = 44.52 \text{ ton-m}$

$a = \dfrac{As\, fy}{0.85 f'c\, b}$ $\quad Mu = \phi As fy (d - \frac{a}{2})$

$0.59\omega^2 - \omega - \dfrac{Mu}{\phi b d^2 f'c} = 0$

$\omega = \rho_d \dfrac{fy}{f'c}$, $\rho_d = \omega \dfrac{f'c}{fy}$, $As_d = \rho_d \, b d$

$\rho_{min} = \dfrac{0.70\sqrt{f'c}}{fy} \Rightarrow \rho_{min} = 0.00242 \longrightarrow As_{min} = \rho_{min} bd \Rightarrow As_{min} = 16.906 \text{ cm}^2$

$\rho_b = 0.85\beta \dfrac{f'c}{fy}\left(\dfrac{6000}{6000+fy}\right) \Rightarrow \rho_b = 0.02125 \longrightarrow As_b = \rho_b bd \Rightarrow As_b = 148.75 \text{ cm}^2$

$\rho_d = \qquad \rho_d = 0.00248 \longrightarrow As_d = \rho_d bd \Rightarrow As_d = 17.332 \text{ cm}^2$

$\rho_{MAX} = 0.75 \rho_b \Rightarrow \rho_{MAX} = 0.01594 \longrightarrow As_{max} = \rho_{max} bd \Rightarrow As_{max} = 111.562 \text{ cm}^2$

$\rho_d < \rho_{MAX} < \rho_b$ ∴ FALLA DUCTIL

DISTRIBUCION DE ACERO: $As_{td} = 17.332(4) \Rightarrow As_{td} = 69.328 \text{ cm}^2$

$As\phi\, 3/4 = 2.84 \text{ cm}^2 \quad d_b = 0.019 \text{ m}$

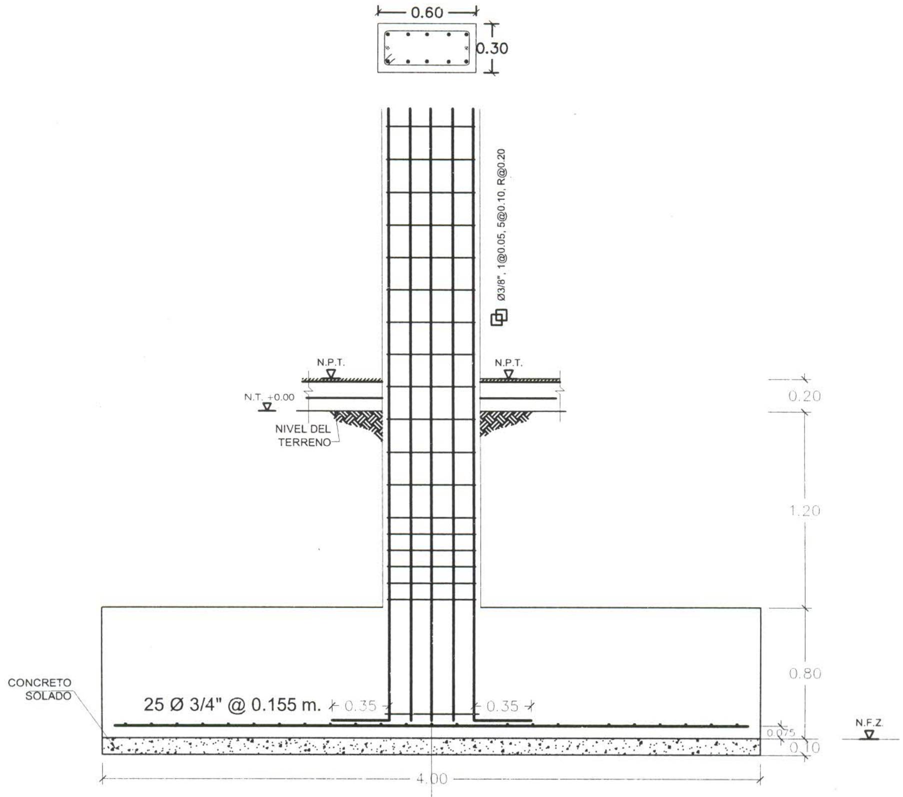
\# ACEROS

$n = \dfrac{As_{td}}{As\phi} \Rightarrow n = \dfrac{69.328}{2.84} \Rightarrow n = 24.411 \Rightarrow n = 25$ UNIDADES.

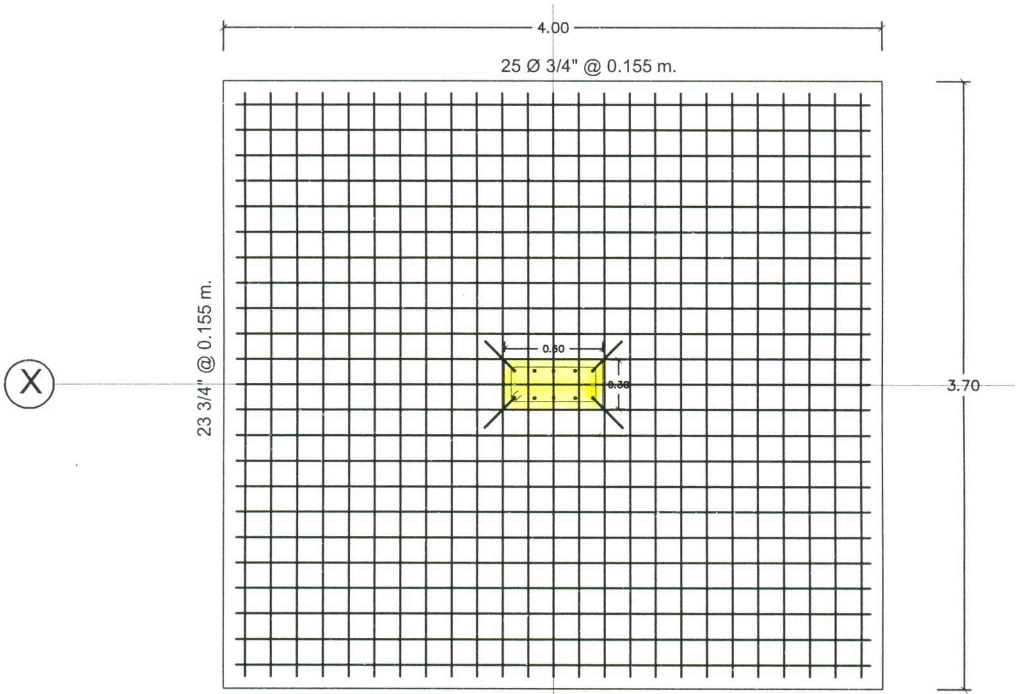
ESPACIAMIENTO:

$s = \dfrac{L - 2r - \phi - 0.10}{n - 1} = \dfrac{4.00 - 0.15 - 0.019 - 0.10}{24} \Rightarrow S = 0.155 \text{ cm}$

$25\phi 3/4 @ 0.155$



DETALLE DE ZAPATA TIPICO
ESCALA 1:25



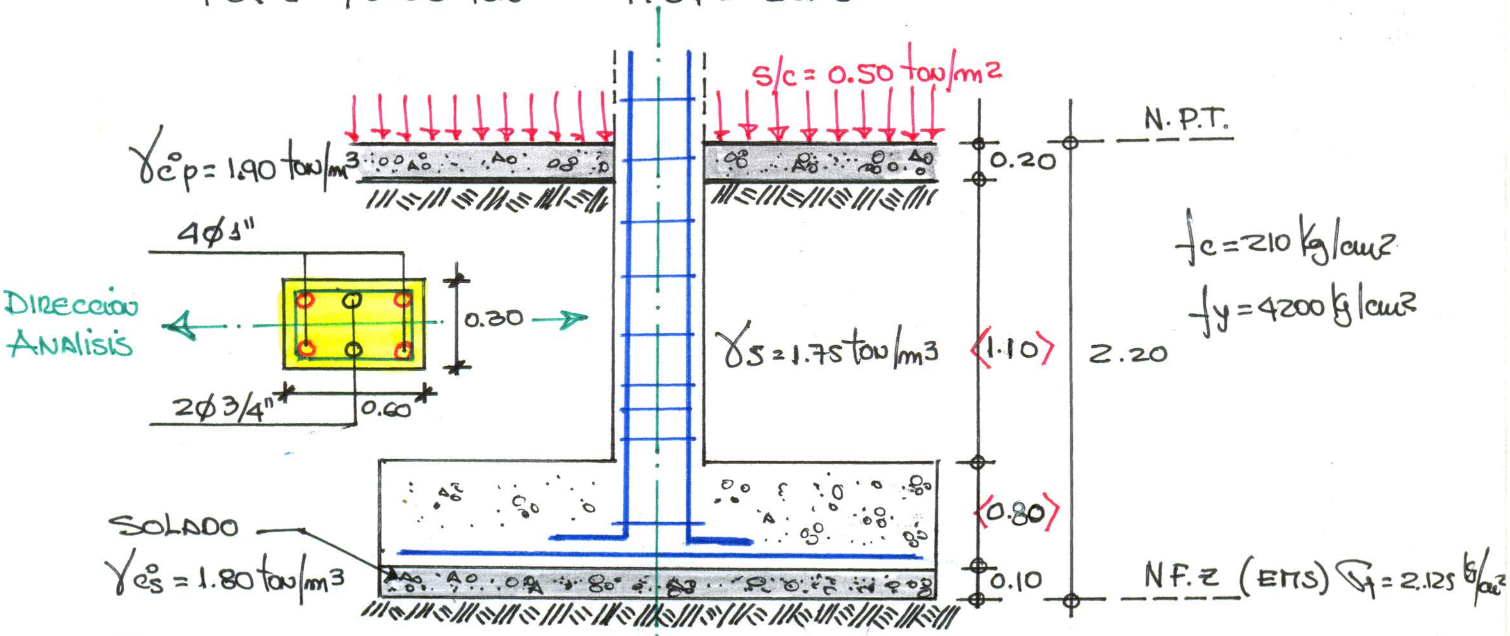
DETALLE DE ZAPATA EN PLANTA
ESCALA 1:25

Y

PROBLEMA #03:

DISEÑAR LA ZAPATA AISLADA CONSIDERANDO LOS DATOS TECNICOS Y ESTRUCTURALES

$P_{CM} = 92.50$ ton $\quad M_{CM} = 36.50$ ton-m $\quad \sigma_t = 2.125$ kg/cm²
$P_{CV} = 73.00$ ton $\quad M_{CV} = 22.50$ ton-m



I): DIMENSIONAMIENTO

PERALTE DE LA ZAPATA: $As_{COL} = 4\phi 1" + 2\phi 3/4 \Rightarrow \phi 1"$ $\quad \begin{cases} db = 2.54 \text{ cm.} \\ Ab = 5.10 \text{ cm2.} \end{cases}$

CONSIDERACIONES NORMATIVAS:

$$\left.\begin{array}{ll} Ld = 0.08\, db\, fy/\sqrt{f'c} & = 58.893 \text{ cm.} \\ Ld = 0.004\, db\, fy & = 42.672 \text{ cm.} \\ Ld \geq 20 \text{ cm.} & = 20.00 \text{ cm} \end{array}\right\} \begin{array}{l} Ld = 58.893 \\ Ld = 60.00 \text{ cm.} \end{array}$$

$\therefore\ d = Ld + 10\text{ cm} \Rightarrow \underline{d = 70.00 \text{ cm.}} \Rightarrow h = d + 10\text{ cm} \Rightarrow \underline{h = 80.00 \text{ cm}}$

CAPACIDAD PORTANTE NETA DEL TERRENO (q_e)

$$q_e = \sigma t - (\gamma_{cs} * h_s) - (\gamma_{cz} * h_z) - (\gamma_s * h_s) - (\gamma_{cp} * h_p) - S/C$$
$$q_e = 21.25 - (1.80 * 0.10) - (2.40 * 0.80) - (1.75 * 1.10) - (1.90 * 0.20) - 0.50$$
$$q_e = 16.345 \text{ ton/m}^2 = 1.6345 \text{ kg/cm}^2$$

AREA DE LA ZAPATA:

$$\boxed{A = \frac{P_T}{q_e}} \Rightarrow A = \frac{P_{CM} + P_{CV}}{q_e} \Rightarrow \frac{92.50 + 73.00}{16.345} \Rightarrow A = 10.125 \text{ m}^2$$

$$A = (t+2m)(b+2m)$$

$$10.125 = (0.60 + 2m)(0.30+2m)$$

$$10.125 = 0.18 + 1.20m + 0.60m + 4m^2$$

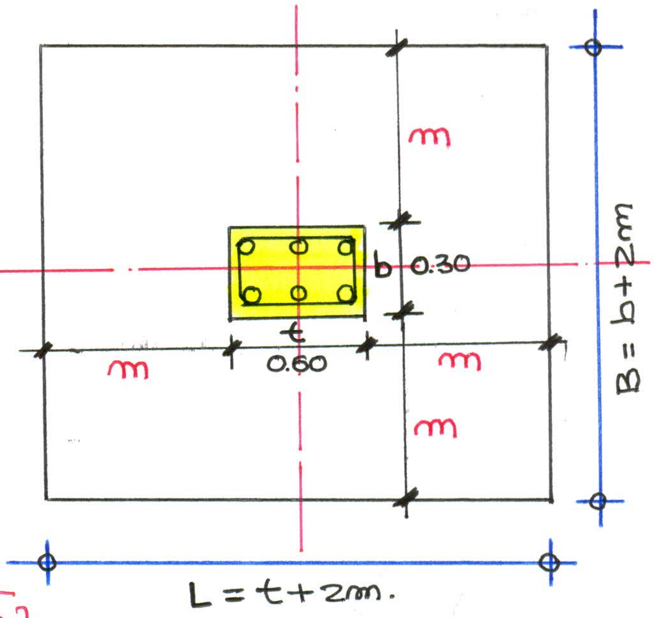
$$4m^2 + 1.8m - 9.945 = \varnothing$$

$$m = 1.368\ m.$$

$$L = t+2m = 0.60 + 2(1.368) \Rightarrow L = 3.336m$$

$$B = b+2m = 0.30+2(1.368) \Rightarrow B = 3.036m$$

$\therefore$ $L = 3.30$, $B = 3.00$, $\underline{A = 9.90\ m^2}$



Verificación de presión $\sigma_{MAX} < \sigma_t$

$$P_s = P_{CM} + P_{CV} = 92.50 + 73.00 \Rightarrow P_s = 165.50\ ton$$

$$M_s = M_{CM} + M_{CV} = 36.50 + 22.50 \Rightarrow M_s = 59.00\ ton\text{-}m$$

$$\boxed{\sigma_{MAX} = \frac{P_s}{A} + \frac{M_s\, c}{I}}$$

$$A = 9.90\ m^2$$

$$c = L/2 \Rightarrow C = 1.65\ m.$$

$$I = \frac{B L^3}{12} \Rightarrow I = 8.984\ m^4$$

$$\sigma_{MAX} = \frac{165.50}{9.90} + \frac{59.00(1.65)}{8.984}$$

$$\sigma_{MAX} = 27.553\ ton/m^2 > \sigma_t = 21.25\ ton/m^2$$

(No cumple la condición)

$\therefore$ Redimensionamos B y L

$$B = 3.40\ m,\quad L = 3.70\ m,\quad A = 12.58\ m^2,\quad c = 1.85m,\quad I = 14.352\ m^4$$

$$\sigma_{MAX} = \frac{P_s}{A} + \frac{M_s * c}{I}$$

$$\sigma_{MAX} = \frac{165.50}{12.58} + \frac{59.00}{14.352}(1.85) \Rightarrow \sigma_{MAX} = 20.761\ ton/m^2$$

$$\therefore \underline{\sigma_{MAX} = 20.761\ ton/m^2} < \underline{\sigma_t = 21.25\ ton/m^2}$$

(CONFORME)

CARGAS DE DISEÑO (P_U, M_U)

REACCIÓN AMPLIFICADA DEL SUELO:

$P_{CM} = 92.50$ ton $M_{CM} = 36.50$ ton-m.

$P_{CV} = 73.00$ ton $M_{CV} = 22.50$ ton-m.

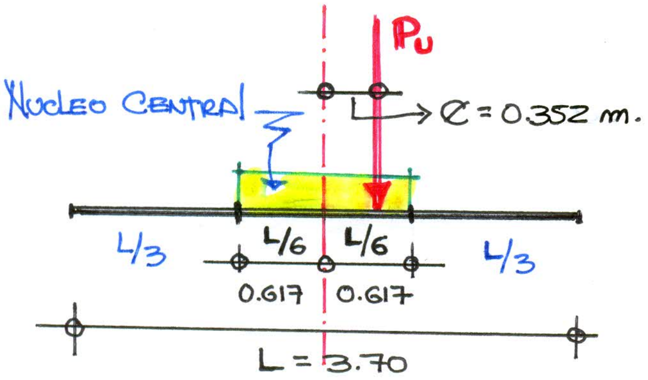
$$\boxed{P_U = 1.70\,P_{CV} + 1.40\,P_{CM}} = 1.70(73.00) + 1.40(92.50) \Rightarrow \underline{253.60 \text{ ton}}$$

$$\boxed{M_U = 1.70\,M_{CV} + 1.40\,M_{CM}} = 1.70(22.50) + 1.40(36.50) \Rightarrow \underline{89.35 \text{ ton-m.}}$$

$$M = F * d \Rightarrow M_U = P_U * e \Rightarrow e = \frac{M_U}{P_U} = \frac{89.35}{253.60} \Rightarrow \underline{e = 0.352\,m.}$$

$$\frac{L}{6} = \frac{3.70}{6} \Rightarrow \frac{L}{6} = 0.617\,m \qquad \therefore\ e = 0.352 < \frac{L}{6} = 0.617$$

PRESIÓN DE CONTACTO FORMA TRAPEZOIDAL



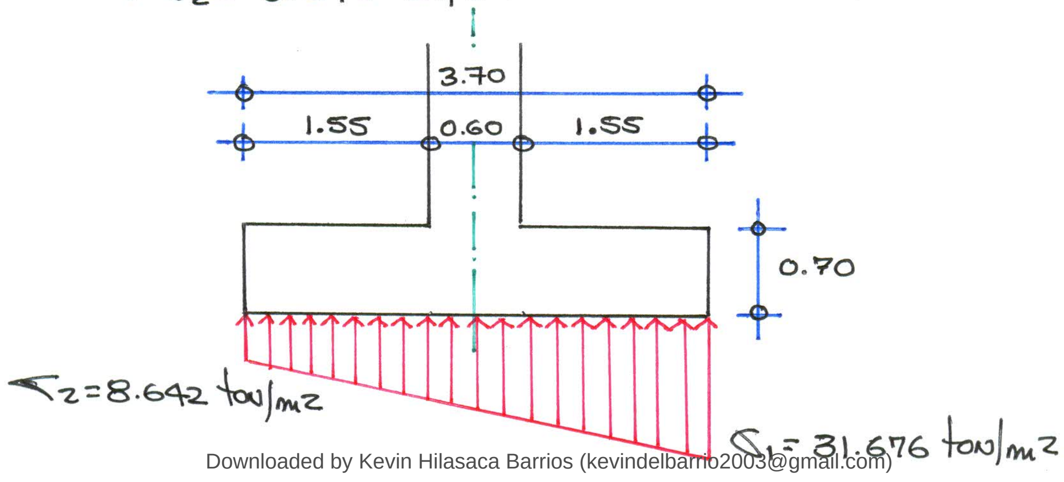
PRESIONES PARA EL DISEÑO (METODO DE RESISTENCIA ULTIMA)

$$\boxed{\sigma_{1,2} = \frac{P_U}{A} \pm \frac{M_U * c}{I}}$$

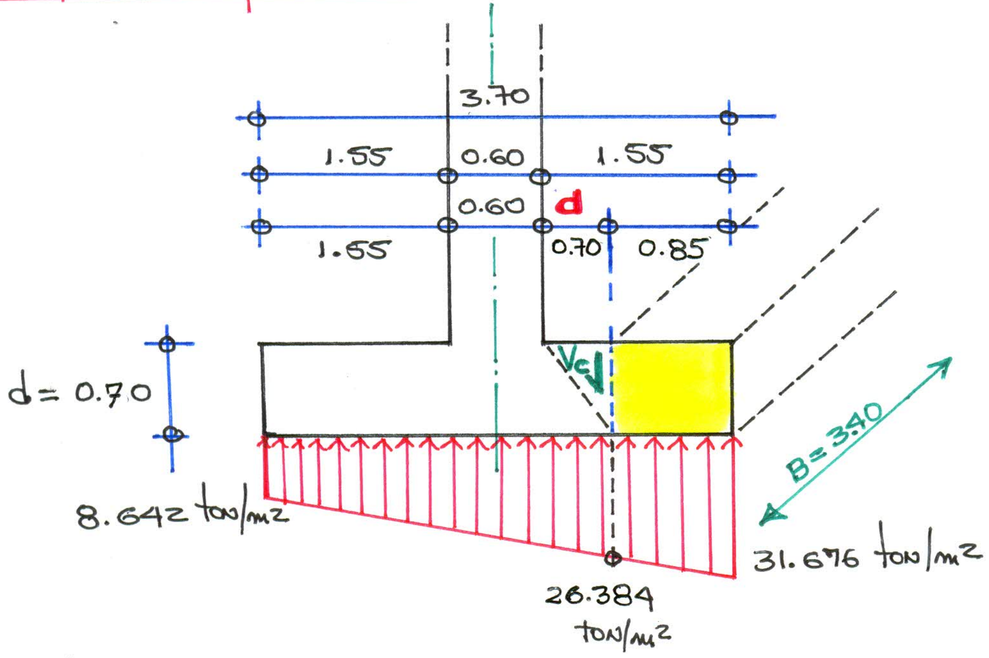
$P_U = 253.60$ ton
$M_U = 89.35$ ton-m.
$L = 3.70$ m
$B = 3.40$ m.
$A = 12.58\ m^2$
$c = 1.85\,m$
$I = 14.352\ m^4$

$$\sigma_{1,2} = \frac{253.60}{12.58} \pm \frac{89.35(1.85)}{14.352}$$

$$\sigma_{1,2} = \begin{cases} \sigma_1 = 31.676 \text{ ton/m}^2 \\ \sigma_2 = 8.642 \text{ ton/m}^2 \end{cases}$$



VERIFICACION POR CORTANTE



$$\boxed{V_u \leq \phi V_c}$$

Fuerza cortante ultima — Resistencia del concreto @ corte

$$V_u = \left[\frac{(31.676 + 26.384)\,0.85}{2}\right] * 3.40$$

$$V_u = 83.897 \text{ ton} = 83896.700 \text{ Kg}$$

$$\phi V_c = \phi\, 0.53 \sqrt{f'c}\; B.d$$

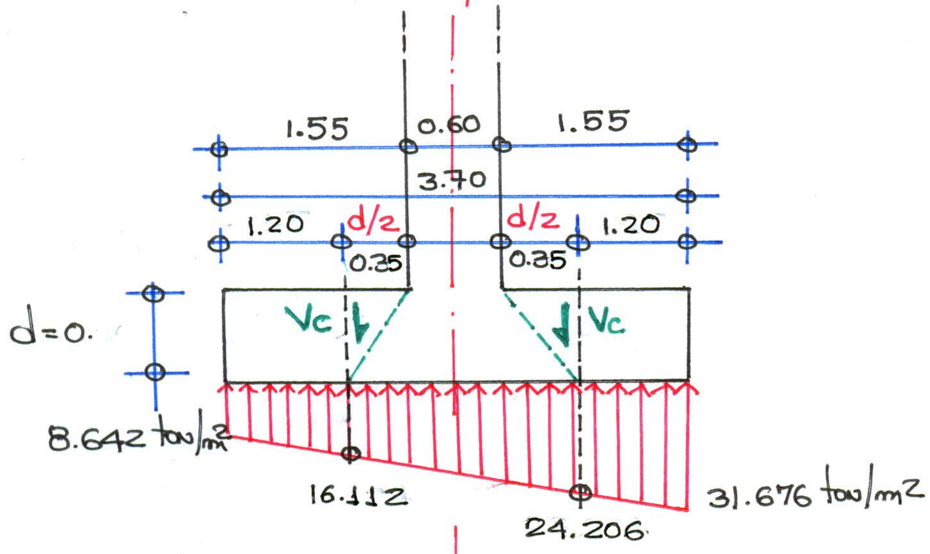
$$\phi V_c = 0.85 * 0.53 \sqrt{210} * 340 * 70$$

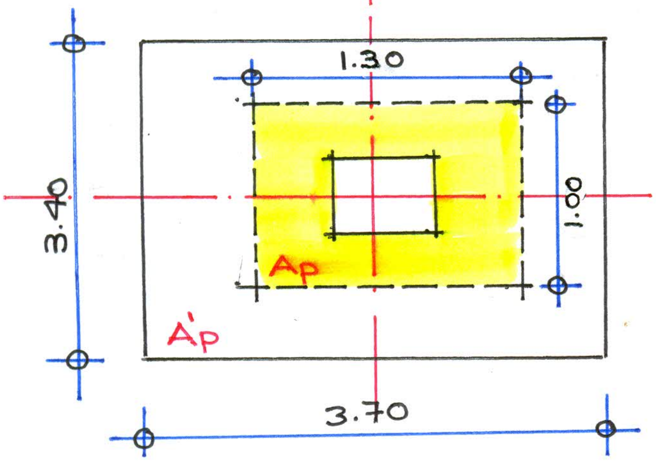
$$\phi V_c = 155.375 \text{ ton} = 155375.092 \text{ Kg}$$

$$\therefore V_u = 83.897 \text{ ton} < \phi V_c = 155.375 \text{ ton}$$

CONFORME

VERIFICACIÓN POR PUNZONAMIENTO





PERIMETRO DE PUNZONAMIENTO

$b_o = 2(1.30) + 2(100) \Rightarrow b_o = 460\ cm$

RELACIÓN LADOS DE COLUMNA

$\beta_o = \dfrac{\text{LADO MAYOR COL.}}{\text{LADO MENOR COL.}}$

$\beta_o = \dfrac{0.60}{0.30} \Rightarrow \beta_o = 2$

$A'p = A_z - A_p$

$A'p = (370 * 340) - (130 * 100)$

$A'p = 112800.000\ cm^2$

$\boxed{V_u \leq \phi V_c}$

Fuerza Cortante Ultima — Resistencia del Concreto @ corte Punzonamiento

$\boxed{V_u = q_u\, A'p}$

$\boxed{\phi V_c = \phi \left(0.53 + \frac{1.10}{\beta_o}\right)\sqrt{f'c}\ b_o\, d}$

$V_u = \left[\dfrac{(31.676 + 8.642)}{2} 3.70\ (3.40)\right] - \left[\dfrac{(24.206 + 16.112)}{2} 1.30 (1.00)\right]$

$V_u = 227.394\ ton = 227393.520\ Kg$

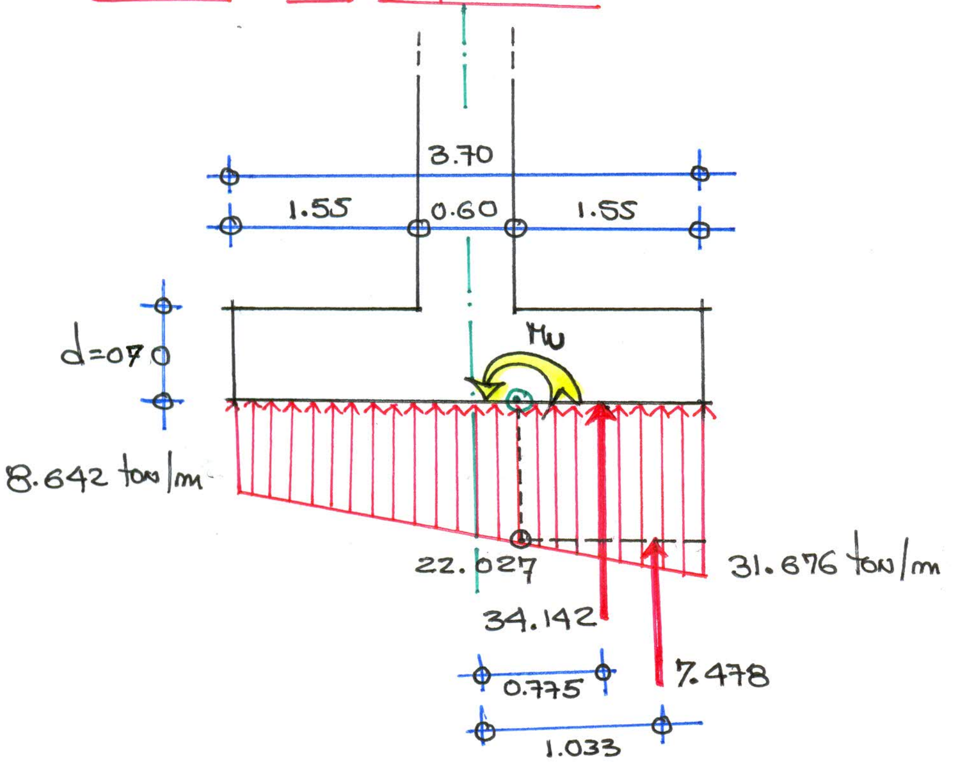
$\phi V_c = 0.85 \left(0.53 + \frac{1.10}{2}\right) \sqrt{210} * 4\ \ 0 * 70$

$\phi V_c = 428.359\ ton = 428.359.300\ Kg.$

$V_u = 227.394\ ton < \phi V_c = 428.359\ ton$

CONFORME

DISEÑO DE REFUERZO:



$Mu = 34.142(0.775) + 7.478(1.033) \Rightarrow Mu = 34.185 \text{ ton-m}$

DISEÑO DE REFUERZO LONGITUDINAL

Datos de Diseño:

$Mu = 34.185$ ton-m, $b = 100$ cm. $d = 70$ cm. $f'c = 210$ kg/cm², $f_y = 4200$ kg/cm²

$$\boxed{a = \frac{As\, f_y}{0.85 f'c\, b}} \qquad \boxed{Mu = \phi As f_y \left(d - \frac{a}{2}\right)}$$

$$\rho_{min} = \frac{0.70\sqrt{f'c}}{f_y} \Rightarrow \rho_{min} = 0.00242 \Rightarrow As_{MIN} = \rho_{min}\, bd \Rightarrow As_{MIN} = 16.906 \text{ cm}^2$$

$$\rho_b = 0.85\beta_1 \frac{f'c}{f_y}\left(\frac{6000}{6000+f_y}\right) \Rightarrow \rho_b = 0.02125 \Rightarrow As_b = \rho_b\, bd \Rightarrow As_b = 148.75 \text{ cm}^2$$

$$\rho_{MAX} = 0.75\rho_b \Rightarrow \rho_{MAX} = 0.01594 \Rightarrow As_{MAX} = \rho_{MAX}\, bd \Rightarrow As_{MAX} = 111.563 \text{ cm}^2$$

$$\rho_b = \omega \frac{f'c}{f_y} \Rightarrow \rho_d = 0.00189 \Rightarrow As_d = \rho_d\, bd \Rightarrow As_d = 13.214 \text{ cm}^2$$

$As\,\phi 3/4 = 2.84 \text{ cm}^2$ $\quad db\,\phi 3/4 = 1.905$ cm. $= 0.01905$ m.

$$As_d < As_{MIN} \Rightarrow As_d = As_{MIN} = 16.906 \text{ cm}^2$$

NUMERO DE VARILLAS $A_{std} = A_{sd}\,(3.40) \Rightarrow A_{std} = 57.480\ cm^2$

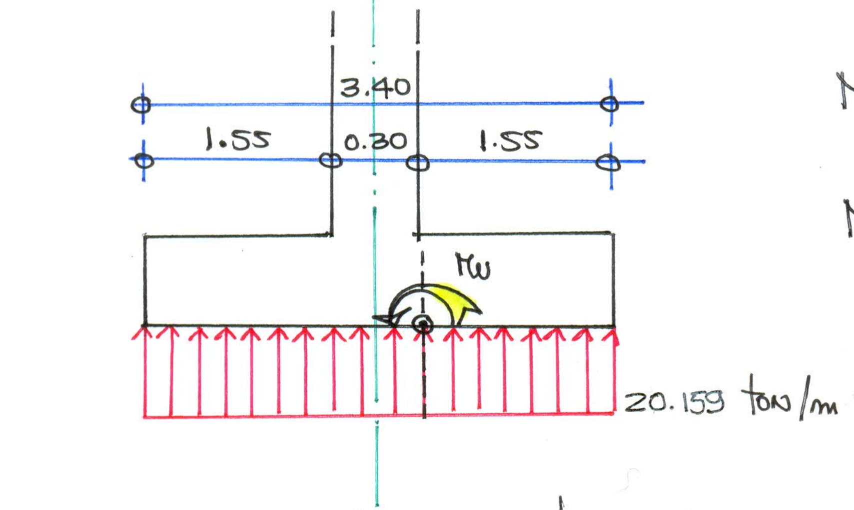
$$N = \frac{A_{std}}{A_{s\phi}} \Rightarrow N = \frac{57.480}{2.84} \Rightarrow N = 20.24 \Rightarrow N = 20 \text{ UNIDADES.}$$

DISTRIBUCION DE ACERO:

$$S = \frac{B - 2r - \phi - 0.10}{N-1} \Rightarrow S = \frac{3.40 - 2(0.075) - 0.01905 - 0.10}{1} \Rightarrow S = 0.165\ m$$

∴ ACERO LONGITUDINAL: $20\phi 3/4$ @ 0.165 m.

DISEÑO DE ACERO TRANSVERSAL:



$$M_u = \frac{qL^2}{2} = \frac{20.159\,(1.55)^2}{2}$$

$$M_u = 24.216\ ton\text{-}m$$

DATOS DE DISEÑO: $M_u = 24.216$ ton-m, $b = 100$ cm. $d = 70$ cm. $f'c = 210\ kg/cm^2$ $f_y = 4200\ kg/cm^2$

$$\rho_{min} = \frac{0.70\sqrt{f'c}}{f_y} \Rightarrow \rho_{min} = 0.00242 \Rightarrow A_{smin} = \rho_{min}\, b d \Rightarrow A_{smin} = 16.90\ cm^2$$

$$\rho_b = 0.85\beta_1 \frac{f'c}{f_y}\left(\frac{6000}{6000+f_y}\right) \quad \rho_b = 0.02125 \Rightarrow A_{sb} = \rho_b\, b d \Rightarrow A_{sb} = 148.75\ cm^2$$

$$\rho_{max} = 0.75\,\rho_b \quad \rho_{max} = 0.01594 \Rightarrow A_{smax} = \rho_{max}\, b d \Rightarrow A_{smax} = 111.56\ cm^2$$

$$\rho_d = \omega \frac{f'c}{f_y} \quad \rho_d = 0.00 \Rightarrow A_{sd} = \rho_d\, b d \Rightarrow A_{sd} = 7.719\ cm^2$$

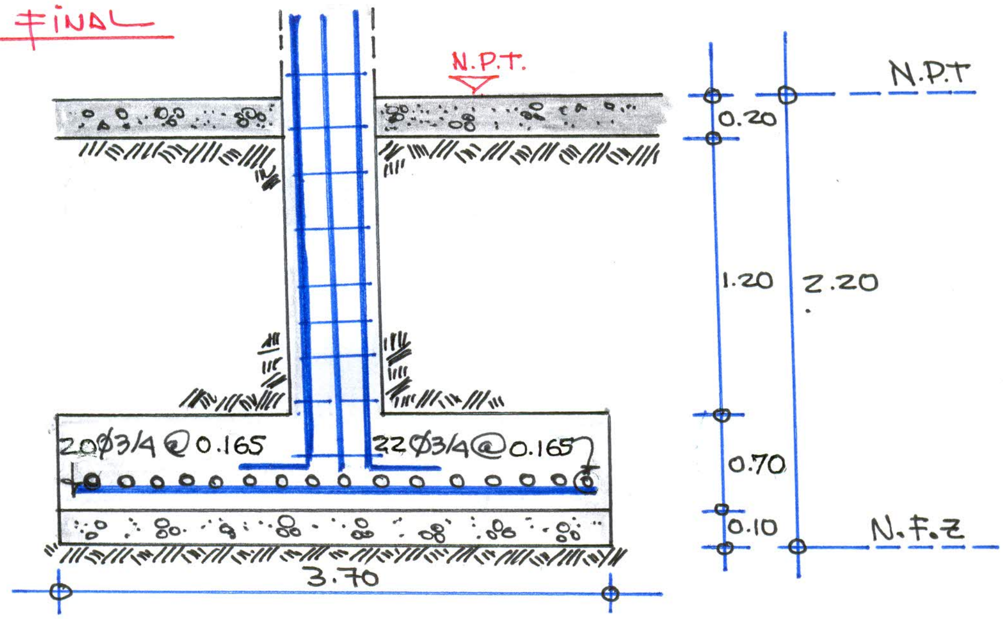
$$A_{sd} < A_{smin} \Rightarrow \underline{A_{sd} = 16.90\ cm^2} \Rightarrow A_{std} = A_{sd}\,(L) \Rightarrow A_{std} = 62.55\ cm^2$$

$$N = \frac{A_{std}}{A_{s\phi}} \Rightarrow N = \frac{62.55}{2.84} \Rightarrow N = 22.027 \text{ UNIDADES} \Rightarrow N = 22 \text{ UNIDADES.}$$

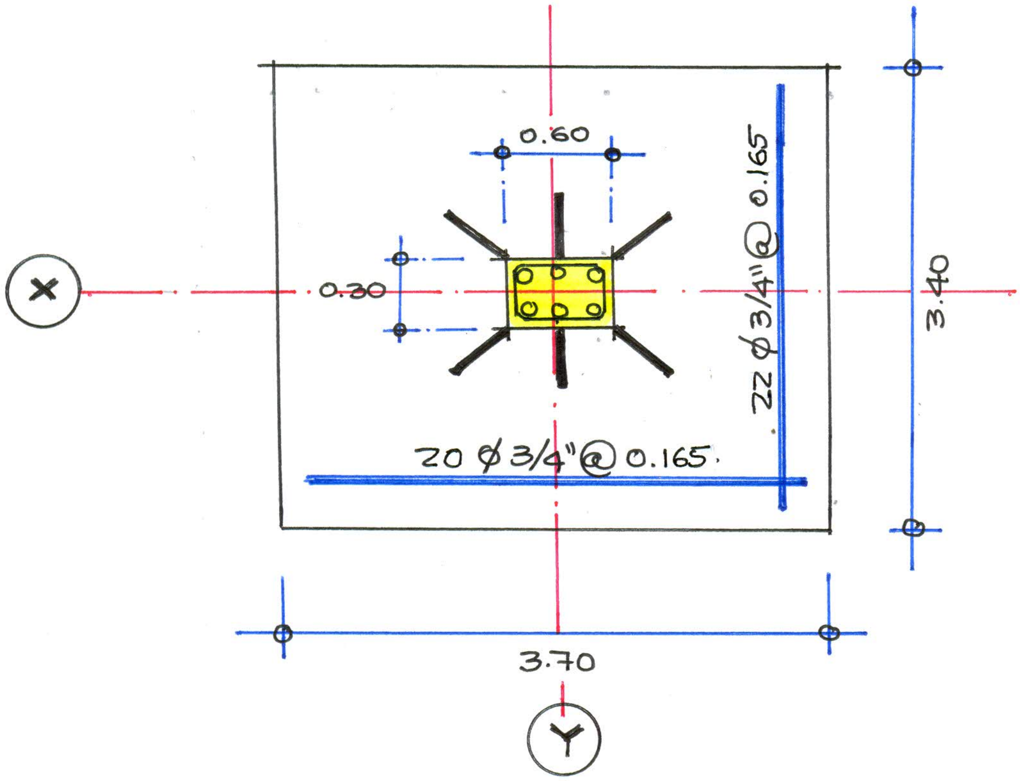
$$S = \frac{L - 2r - \phi - 0.10}{N-1} \Rightarrow \frac{3.70 - 0.15 - 0.01905 - 0.10}{21} \Rightarrow S = 0.16\ m.$$

∴ ACERO TRANSVERSAL: $22\phi 3/4"$ @ 0.16 m

DISEÑO FINAL



SECCION : X–X



PLANTA TIPICO Z-01

DETALLE FINAL DE ZAPATA